TMZ

TAMIZHI BLOCKCHAIN

Tamil Cultural Heritage on a Cosmos SDK L1

Project Status Report

A Cosmos SDK L1 chain preserving Tamil cultural heritage

17/17 Test packages	**0** Build errors	**100%** Code complete
11 Chain modules	**4** Rust contracts	**53** Tamil works

Report Date: May 04, 2026
Mainnet Genesis: June 1, 2026

Executive Summary

Tamizhi is a production-ready Cosmos SDK Layer-1 blockchain dedicated to preserving Tamil literary heritage, powering employee rewards, secure messaging, and cultural NFT ownership. The technical build is **100% complete** — every chain module, smart contract, frontend, and service has been written, tested, and verified.

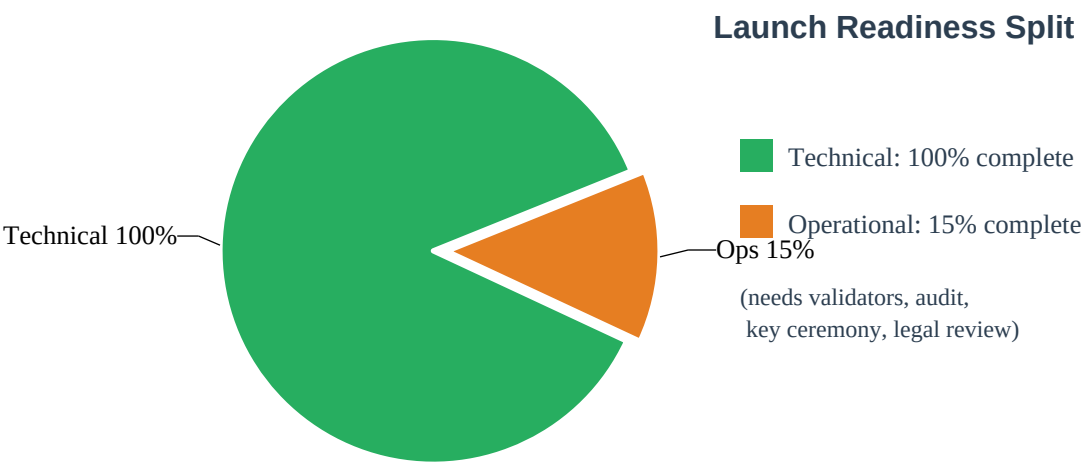
All 17 Go test packages pass with zero failures. The full **go build ./...** succeeds cleanly after the wasmd v0.53.4 upgrade. CosmWasm contracts compile, the TypeScript SDK type-checks, and the entire Docker stack starts without manual intervention.

Remaining work to mainnet is **operational, not technical**: validator recruitment, the 3-of-5 multisig key ceremony, an external security audit, and legal review of token distribution. Genesis is scheduled for **June 1, 2026**.

Launch Readiness

Technical Readiness 100%

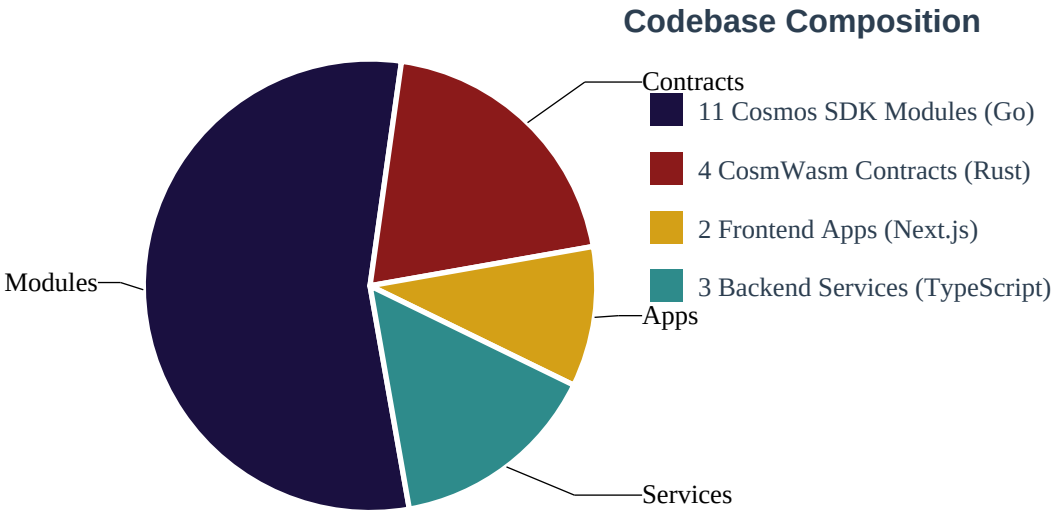
Operational Readiness 15%



System Architecture

Six layers from consensus to user-facing apps. Each layer is independently deployable, tested, and documented.

Codebase Composition



Component Status

Chain Layer

Component	Detail	Statu
Cosmos SDK	v0.50.9 with CometBFT BFT consensus	Read
Custom modules	rewards, nodes, accesslog, security, messaging, identity, literature, nft, feeburn, ibcmiddleware, wasm	11 / 1
Custom ante chain	Blocklist + Rate-limit + MinFee + FeeBurn (20%)	Read
IBC integration	ICS-20 + AccessLog middleware (auto-logs incoming transfers)	Read
CosmWasm VM	v0.53.4, 4 Rust contracts compiled & tested	Read
Governance	MsgUpdateParams for 4 modules + 3-of-5 multisig admin	Read
Upgrade handler	v1.1.0 with pre/post hooks, Cosmovisor compatible	Read

Application & Service Layer

Component	Tech Stack	Status
Wallet frontend	Next.js 14 + Keplr/Leap + CosmJS	Ready
Literary browser	Next.js 14 + Tamil typography (Noto Sans Tamil)	Ready
Block explorer	Node.js + PostgreSQL indexer + Express REST API	Ready
WebSocket relay	Tendermint WS → browser event streaming	Ready
Discord bot	discord.js — 7 slash commands + faucet + alerts	Ready
Telegram bot	telegraf — 7 commands + chain stats	Ready
TypeScript SDK	@tamizhi/chain-sdk — REST + signing + gRPC-Web	Ready
Envoy gRPC-Web	Browser-compatible gRPC proxy	Ready

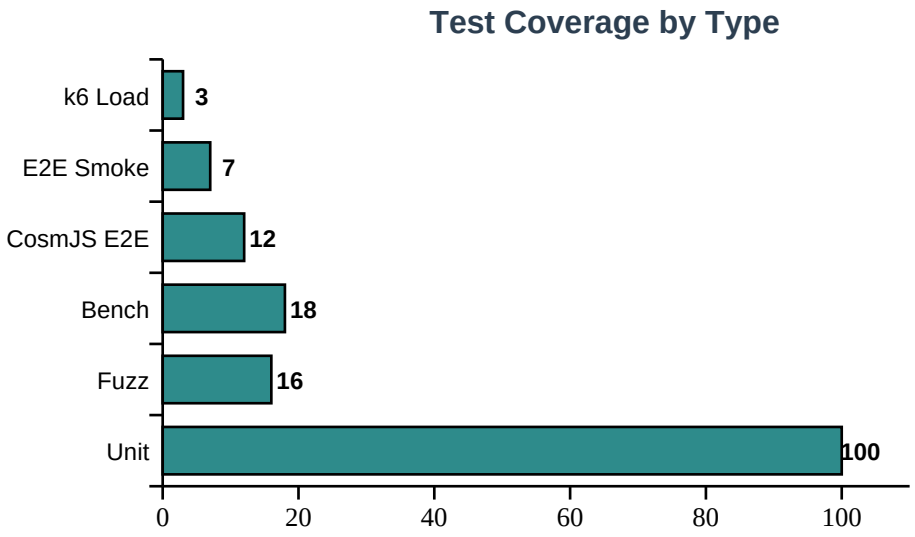
Infrastructure & DevOps

Component	Detail	Status
Docker localnet	2 validators + seed, P2P-peered	Ready
Helm chart	Kubernetes deployment manifests	Ready
GitHub Actions	3 workflows: CI, fuzz (weekly), release	Ready
Prometheus + Alertmanager	14 alert rules across 5 categories	Ready
Grafana dashboard	10 panels live + provisioned	Ready
Cosmovisor	Zero-downtime upgrade orchestration	Ready
State sync	Snapshot/restore for fast bootstrap	Ready

Component	Detail	Status
k6 load tests	3 scenarios: write stress, read-heavy, spike	Ready

Quality & Test Coverage

Testing spans every layer of the stack — from message validation fuzzing to multi-validator end-to-end scenarios. CI runs the full suite on every push. Fuzz tests run weekly on a schedule.



Test Type	Count	Coverage
Unit tests	100+	All chain modules
Fuzz functions	16	All Msg.ValidateBasic across 5 modules
Benchmarks	18	Hot paths in rewards, nodes, accesslog, security
Integration	12	CosmJS sign + broadcast against localnet
E2E smoke	7 cases	Full localnet lifecycle
Load (k6)	3 scenarios	Up to 200 VUs spike
Contract tests	16	9 literary-nft + 7 rewards escrow

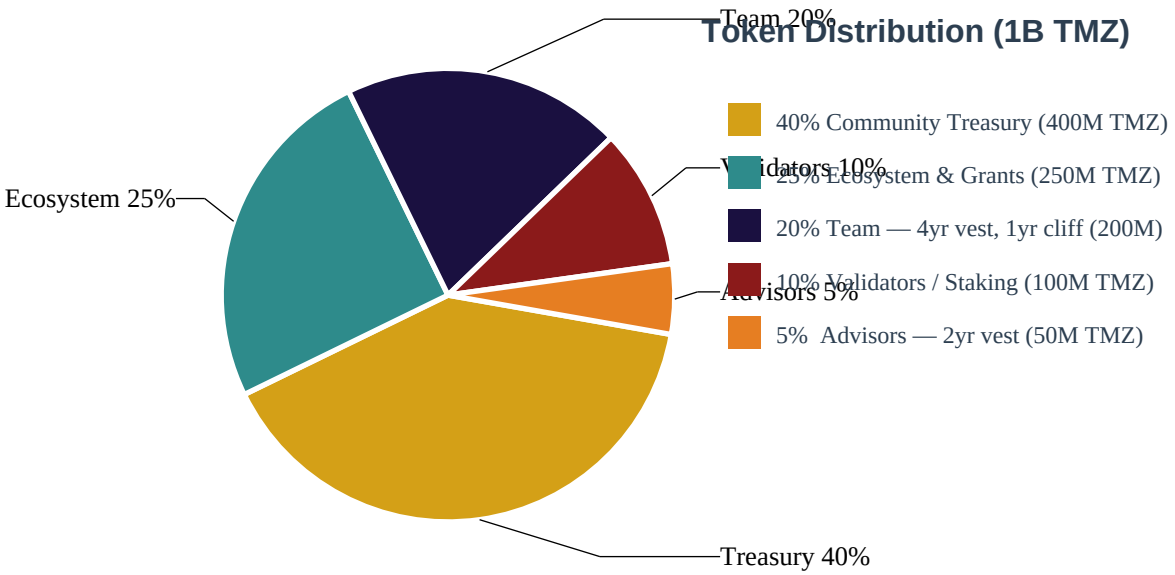
Security Posture

Control	Implementation
Address blocklist	On-chain via x/security; ante decorator rejects flagged addresses
Rate limiting	50 tx/block/address — protects against spam
Minimum fee	1,000 utmz — anti-spam economic deterrent
Fee burn	20% of all fees burned — deflationary pressure
Node slashing	5% stake slash + jail when UptimeScore < 20
Double-claim prevention	CosmWasm rewards escrow uses (employee, rule) composite key
Multisig admin	3-of-5 signing required for governance actions

Control	Implementation
Validator key protection	tmkms remote signer recommended in production

Tokenomics — TMZ

TMZ is the native staking and governance token. Total supply is fixed at **1 billion TMZ** (1,000,000,000,000,000 utmz). The base unit is **utmz**; 1 TMZ = 1,000,000 utmz.

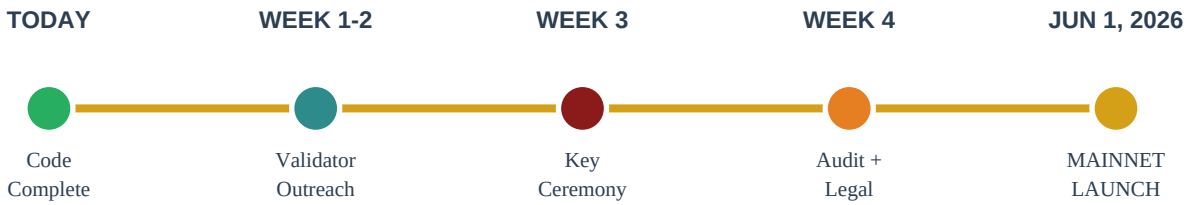


Fee Economics

Parameter	Value	Note
Minimum fee	1,000 utmz / tx	Enforced by ante handler
Burn rate	20% of fees	Sent to no-key address
Validator share	72% of fees	After burn + community
Community pool	8% of fees	Funds grants and ecosystem
Block target	5 seconds	CometBFT BFT consensus
Min validator stake	15,000 TMZ	Slash-protected via tmkms
Unbonding period	14 days	Standard Cosmos default

Roadmap to Mainnet

Path to Mainnet — June 1, 2026



Phase	Item	Owner	ETA
Week 1-2	Recruit 4+ external validators	Community	Q2 2026
Week 1-2	Deploy public testnet RPC + state-sync service	DevOps	1 day
Week 2	Run 24h fuzz test (background)	CI	Automated
Week 3	3-of-5 multisig key ceremony (offline)	5 keyholders	Half day
Week 3	Wire multisig address into mainnet genesis	Lead Dev	1 day
Week 3-4	External security audit	Audit firm	2-4 weeks
Week 3-4	Legal review of token distribution	Counsel	1-2 weeks
Week 4	Validator dry-run on testnet	All	1 week
June 1, 2026	Mainnet genesis (00:00:00 UTC)	All	Launch day
Post-launch	Mobile wallet (React Native)	Frontend	Q3 2026

What's Left Before Launch

Every item below is **operational**, not technical. The codebase requires no further changes before mainnet.

Priority	Action Item	Status
P0	Recruit 4+ independent validators	Pending
P0	3-of-5 multisig key ceremony	Pending
P0	External security audit	Pending
P0	Legal review of tokenomics	Pending
P1	Deploy public testnet RPC	Pending
P1	Deploy monitoring stack (Prom/Grafana)	Pending
P1	Create Discord/Telegram bot tokens	Pending
P1	Fund faucet wallet	Pending
P2	Validator dry-run on testnet	Pending
P2	Final smoke test (e2e-smoke-test.sh)	Pending

Immediate Actions This Week

• Post validator recruitment in the Cosmos Discord **#validator-lounge** channel

• Send direct messages to 10 known validators (Lavender.Five, Polkachu, Notional, KalpaTech, etc.)

• Engage one of three audit firms: Oak Security · Informal Systems · Halborn

• Identify and brief the 5 keyholders for the multisig ceremony

• Spin up a testnet VPS (4 CPU, 16GB RAM, 500GB SSD) for the public RPC

• Run **make launch-check** weekly to track progress

Project Resources

Resource	Location
Repository	github.com/tamizhi/chain (private)
Chain ID (testnet)	tamizhi-testnet-1
Chain ID (mainnet)	tamizhi-1
Native denom	utmz (1 TMZ = 1,000,000 utmz)
Block explorer	explorer.tamizhi.ai (post-launch)
Public RPC	rpc.tamizhi.ai (post-launch)
REST API	api.tamizhi.ai (post-launch)
Documentation	docs.tamizhi.ai (post-launch)
Discord	discord.gg/tamizhi (TBD)
Telegram	t.me/TamizhiChain (TBD)
Security contact	security@tamizhi.ai

Key Documentation Files

File	Description
LAUNCH_CHECKLIST.md	Tier-1/2/3 go/no-go checklist with launch-day runbook
docs/architecture.md	Full technical architecture with module dependency graph
docs/tokenomics.md	Token economics, vesting schedule, governance model
docs/key-ceremony.md	Step-by-step 3-of-5 multisig ceremony procedure
docs/validator-security.md	Validator hardening, sentry architecture, tmkms
docs/runbook.md	Six incident response procedures with diagnosis commands
SECURITY_AUDIT.md	Internal audit report covering all 10 fixed bugs

Tamizhi Blockchain

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Genesis: June 1, 2026 — 00:00:00 UTC